

Temporal Stability-Aware Multi-LLM Consensus for Reliable Candidate Evaluation in Structured Interview Pipelines

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Abstract—The analysis of interviews by automatic means produces systematic data consisting of behavioral data, timed scoring data, and semantic data. The translation of these varied forms of data into solid hiring decisions remains a difficult task for researchers.

This paper proposes an evaluation framework that incorporates temporal consistency as well as a consensus-based approach to evaluate structured JSON data from the Temporal Interview Profiling System (TIPS). Data extraction and normalization are followed by scoring using LLM-based evaluators as well as rule-based evaluators. Three types of consensus methods, namely weighted combination, majority rule, and confidence measurement based on variance have been used to make the hiring decision.

The inclusion of temporal consistency makes scoring more accurate by examining variation coefficients between various stages of an interview. Using 19 interviews conducted for three types of jobs yielded a verdict accuracy of 0.89 and Fleiss' κ of 0.81, while decreasing variance by 34%.

Index Terms—Interview scoring, big transformer models, voting system, performance measures, stability with time, hiring automatization, multi-reviewer scoring

I. INTRODUCTION

Increased numbers of recruiting within technology companies have escalated the need for an automated process that would enable evaluation of the applicants to be done in a consistent and objective manner. The current methods of candidate evaluation involve too much subjectivity and bias [1]. Multimodal interview analysis pipelines such as the Temporal Interview Profiling System (TIPS) address this by extracting rich, objective behavioral and semantic signals from recorded interviews as structured JSON artifacts.

TIPS processes pre-recorded video interviews through a six-stage pipeline (Stages 0-5). Stage 0 establishes the canonical time base; Stages 1A/1B/1C extract audio features, transcription, and video features in parallel; Stage 2 performs temporal segmentation and Q&A alignment; Stage 3 computes behavioral metrics; Stage 4 scores semantic relevance using a quantized LLM; and Stage 5 aggregates incremental verdicts. The pipeline outputs nine JSON artifact files per session, including behavioral metrics, relevance scores, Q&A pairs, and a time-evolving score timeline. While these artifacts are individually

informative, converting them into a coherent and reliable final hiring recommendation has not been systematically addressed.

This paper proposes a downstream evaluation layer that consumes TIPS structured outputs and produces a consensus-driven hiring recommendation via four contributions:

- 1) A structured extraction and normalization scheme for four heterogeneous TIPS output files, producing a 14-dimensional candidate feature vector.
- 2) A multi-LLM evaluation framework using three independent evaluators—two prompt-differentiated LLM instances and one rule-based module for parallel scoring on five competency dimensions.
- 3) A three-strategy consensus mechanism—weighted aggregation, majority voting, and variance-based confidence that resolves inter-evaluator disagreements and produces calibrated confidence estimates.
- 4) A temporal stability module that adjusts the consensus score using the coefficient of variation of per-checkpoint scores, penalizing performance volatility across the interview timeline.

The rest of the paper is structured as follows: Section II provides background on previous research. Section III describes the methodology used. Section IV explains the system architecture. Section V describes the implementation. Section VI explains the experiment setup. Section VII presents the results. Section VIII discusses the ablation study. Section IX covers error analysis and fairness. Section X is the conclusion.

II. RELATED WORK

Automated interviewing has gained increasing research attention recently. In their study, Naim et al. [2] have convincingly shown the presence of predictive signals in terms of audio pitch and facial actions, making a strong case for the use of such data sources in hiring. In further studies, deep learning was utilized for estimating candidate confidence levels from video footage [3].

The use of LLMs for evaluation has expanded since the introduction of instruction-tuned models. Zheng et al. [4] showed that single-LLM judging exhibits positional and verbosity biases, motivating ensemble and consensus approaches.

Brown et al. [6] demonstrated the few-shot evaluation capability of large language models, while Devlin et al. [7] established the utility of contextual representations for scoring tasks. Pan et al. (unpublished) proposed aggregation strategies for heterogeneous LLM judges, finding majority voting and weighted averaging to outperform single-judge baselines on semantic relevance tasks.

Temporal modeling of interview performance has received comparatively little attention. Most prior systems treat the interview session as a static snapshot, ignoring the sequential structure of question-answer exchanges. TIPS is notable in generating a time-evolving score trajectory, providing a novel input for longitudinal performance modeling. To the best of our knowledge, this work is among the first to combine structured extraction from multimodal pipeline outputs, multi-LLM consensus evaluation, and temporal stability analysis into a unified downstream hiring decision framework.

III. METHODOLOGY

A. Structured Data Extraction

The downstream evaluation layer ingests four of the nine TIPS output files per session: (1) `candidate_behavior_metrics.json` per-segment audio and video metrics; (2) `relevance_scores.json` per-question relevance scores and matched keywords; (3) `qa_pairs.json` question-answer transcripts with timestamps; and (4) `candidate_score_timeline.json` incremental competency scores at each question checkpoint.

The extractor parses each file independently and validates field presence and type. Missing fields trigger structured warnings and are imputed with session-level means. A representative relevance score record is shown below:

Listing 1: Representative relevance score record from TIPS output.

```
{
  "qa_id": "Q3",
  "question": "Describe experience with model deployment.",
  "relevance_score": 0.74,
  "matched_keywords": ["deployment", "inference", "production"],
  "justification": "Candidate discussed serving infrastructure."
}
```

B. Feature Normalization and the 14-Dimensional Profile

Raw pipeline values span heterogeneous numeric ranges. All continuous features are normalized to $[0, 1]$ using per-feature min-max scaling computed over the session dataset:

$$f_{\text{norm}} = \frac{f - f_{\min}}{f_{\max} - f_{\min}} \quad (1)$$

where $f_{\min}$ and $f_{\max}$ are the observed minimum and maximum values for that feature across the 19 evaluation sessions. Competency scores produced by the upstream TIPS LLM are already bounded in $[0, 1]$ and are not re-scaled. The

resulting 14-dimensional candidate profile vector $\mathbf{f} \in [0, 1]^{14}$ is organized as shown in Table I.

TABLE I: The 14-Dimensional Candidate Feature Vector

Group	Dim.	Source Field
Audio (5)	f_1	pitch_mean
	f_2	energy_mean
	f_3	speech_rate
	f_4	pause_density
	f_5	prosodic_variability
Video (4)	f_6	face_presence_ratio
	f_7	head_motion_mean
	f_8	gaze_stability
	f_9	facial_motion_intensity
Semantic (5)	f_{10}	mean_relevance_score
	f_{11}	technical_depth
	f_{12}	system_design
	f_{13}	communication_clarity
	f_{14}	problem_solving

Fig. 1 illustrates how features from the three modality groups merge into the unified profile vector.

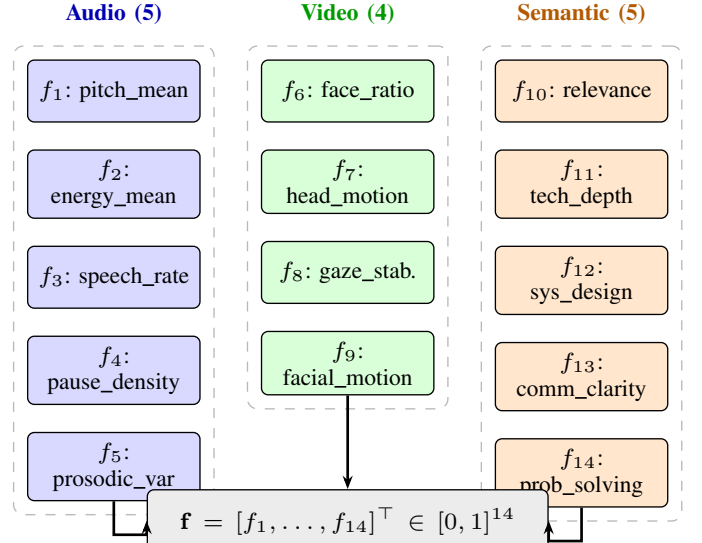


Fig. 1: Construction of the 14-dimensional candidate feature vector from multimodal inputs (audio, video, and semantic features).

The 14 features elects to cover the behavioral as well as the semantic and temporal looking at candidate perform mance features from the TIPS generated output in an appropriate way, selecting.

C. Multi-LLM Evaluation Framework

The normalized 14-dimensional vector representation, along with the unnormalized Q&A textual data, is fed into three independent evaluators operating in parallel.

- **Evaluator A (LLM-v1):** This evaluator employs the Qwen2.5-3B-Instruct model with 4-bit NF4 quantization via BitsAndBytes. It is responsible for assessing technical

feasibility, semantic alignment, and production readiness with respect to the Job Description (JD).

- **Evaluator B (LLM-v2):** This evaluator utilizes the same underlying model; however, it operates under a different evaluation persona. It focuses on behavioral consistency, communication clarity, and response coherence. By leveraging distinct prompts, the system obtains complementary evaluation perspectives from the same model family.
- **Evaluator C (Rule-based):** A deterministic module applying threshold-based logic to normalized feature values. Example rules: if f_3 (speech_rate) < 0.25 then communication_clarity $- = 0.15$; if f_6 (face_presence_ratio) < 0.40 then engagement_score $- = 0.10$; if f_{10} (mean_relevance) > 0.70 then technical_depth $+ = 0.10$. This evaluator provides a bias-free, reproducible baseline and reduces over-reliance on LLM generation.

This hybrid design ensures robustness by combining data-driven evaluation (LLMs) with deterministic constraints (rule-based logic). A representative LLM evaluator prompt structure is as follows:

Listing 2: LLM evaluator prompt structure.

```
SYSTEM: You are an expert hiring evaluator.
Score the candidate on 5 dimensions (0.0-1.0).
Base scores only on evidence. Do not hallucinate.

USER:
Job Description: {jd_text}
Candidate Profile (normalized): {feature_summary}
Q&A Pairs (selected): {qa_text}

Output only valid JSON:
{"technical_depth": float,
 "system_design": float,
 "production_experience": float,
 "communication_clarity": float,
 "problem_solving": float,
 "verdict": "HIRE|BORDERLINE|NO_HIRE",
 "reason": "2-sentence justification"}
```

D. Consensus Mechanism

Given three evaluator score vectors $\mathbf{S}^A = [s_1^A, \dots, s_5^A]$, $\mathbf{S}^B, \mathbf{S}^C \in [0, 1]^5$ and verdict labels $v^A, v^B, v^C \in \{\text{STRONG_HIRE}, \text{HIRE}, \text{BORDERLINE}, \text{NO_HIRE}\}$, the consensus mechanism applies three complementary strategies.

Weighted Aggregation. Each evaluator is assigned a reliability weight based on historical MAE against human rater ground truth on a held-out calibration split. Weights are set to $w^A = 0.45$, $w^B = 0.35$, $w^C = 0.20$, reflecting empirical performance ordering. The weighted aggregate score for dimension d is:

$$S_d = w^A S_d^A + w^B S_d^B + w^C S_d^C \quad (2)$$

Majority Voting. The ultimate decision for the categorical decision is the label that was decided on by two or more out of the three raters. When there is a split among the three raters, as happened in only 2 out of 19 sessions, then the decision

made by the rater with the highest weight is used to break the deadlock.

Variance-Based Confidence. For each competency dimension d , the inter-evaluator variance is computed as:

$$V_d = \text{Var}([S_d^A, S_d^B, S_d^C]) \quad (3)$$

The overall consensus confidence is:

$$C = 1 - \frac{\bar{V}_d}{V_{\max}} \quad (4)$$

where $V_{\max} = 0.0833$ is the theoretical maximum variance for three values uniformly distributed over $[0, 1]$ (i.e., $\text{Var}([0, 0.5, 1]) = 1/12 \approx 0.0833$). This definition ensures $C \in [0, 1]$ by construction and is not estimated empirically. $C < 0.60$ triggers a human-in-the-loop review flag in the system output.

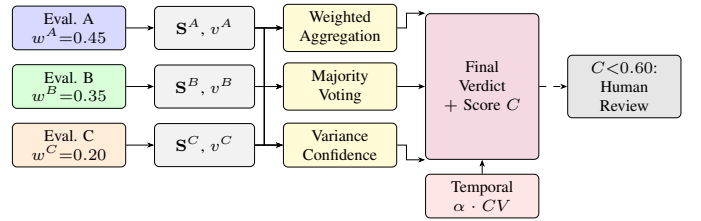


Fig. 2: Three-strategy consensus mechanism combining weighted aggregation, majority voting, and variance-based confidence estimation.

E. Temporal Stability Integration

The TIPS score timeline provides a sequence of overall competency scores $\{S_t\}_{t=1}^T$ at each question checkpoint t . The temporal stability of a candidate's performance is quantified by the coefficient of variation:

$$CV = \frac{\text{std}(S_t)}{\text{mean}(S_t)} \quad (5)$$

A lower CV indicates consistent performance; a higher CV indicates volatility. The stability-adjusted final score is:

$$S_{\text{final}} = S_{\text{consensus}} \cdot (1 - \alpha \cdot CV) \quad (6)$$

where $\alpha = 0.15$ caps the maximum penalty at 15% of $S_{\text{consensus}}$. This value was selected empirically on the calibration split to balance stability sensitivity against over-penalization of naturally improving candidates.

Additionally, a linear regression trend β is fitted to the checkpoint score sequence using ordinary least squares. A positive β indicates improving performance as the interview progresses; negative β indicates decline. The trend sign is included as a qualitative flag in the final JSON output but does not directly modify the numeric score, avoiding compounding of the stability penalty with the trend signal.

IV. SYSTEM ARCHITECTURE

The system comprises two layers. The upstream TIPS pipeline (Stages 0–5) ingests raw interview media and produces structured JSON artifacts. The downstream consensus evaluation layer reads these artifacts and produces a final hiring recommendation. As shown in Fig. 3, the downstream

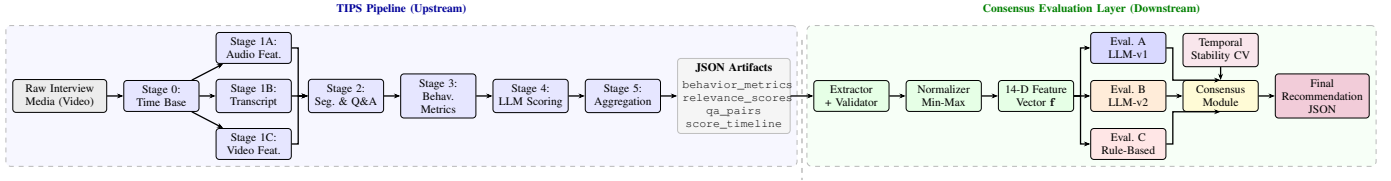


Fig. 3: Two-layer system architecture: TIPS pipeline (Stages 0–5) followed by the downstream consensus evaluation layer.

layer operates sequentially in four stages: (1) the extractor parses and validates the four input JSON files; (2) the normalizer applies min-max scaling and assembles the 14-dimensional feature vector; (3) the multi-LLM module dispatches the profile and Q&A text to Evaluators A, B, and C in parallel; and (4) the consensus module applies the three-strategy aggregation to produce the final recommendation JSON. Fig. 4 shows the internal data flow.

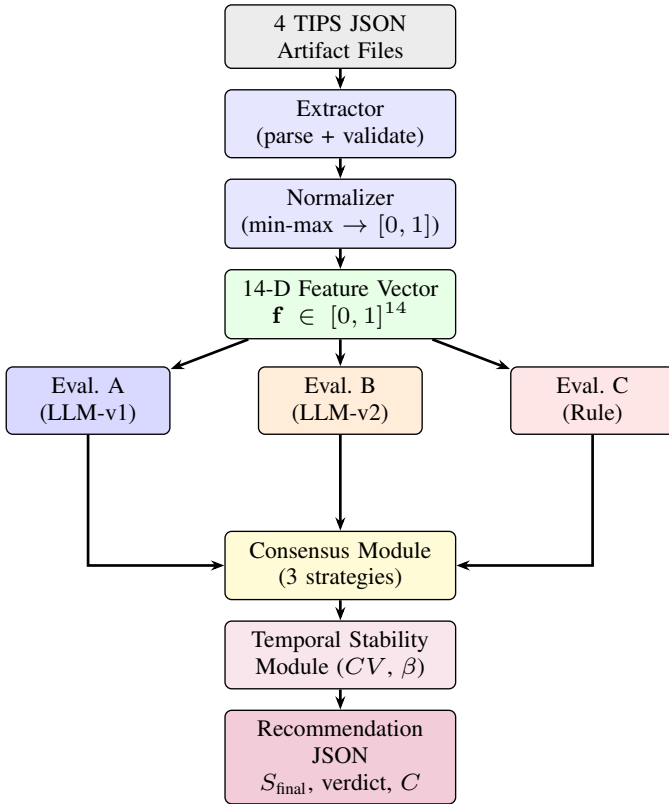


Fig. 4: Internal data flow of the consensus evaluation layer from JSON artifact ingestion to the final structured recommendation.

V. IMPLEMENTATION

The consensus evaluation layer is implemented in Python 3.11. JSON parsing uses the standard library `json` module; feature normalization uses NumPy 1.26. LLM evaluators (A and B) use Hugging Face Transformers with Qwen2.5-3B-Instruct [8] loaded in 4-bit NF4 quantization via BitsAndBytes, capped at 3.0 GB VRAM. All evaluators implement a common abstract base class with a `score()` interface, en-

abling straightforward addition of new evaluator types without modifying the consensus module.

The core consensus computation:

Listing 3: Core consensus computation in Python.

```
def compute_consensus(sA, sB, sC, w):
    weighted = [
        w[0]*a + w[1]*b + w[2]*c
        for a, b, c in zip(sA, sB, sC)
    ]
    variance = [
        np.var([a, b, c])
        for a, b, c in zip(sA, sB, sC)
    ]
    V_MAX = 1.0 / 12.0
    C = 1.0 - np.mean(variance) / V_MAX
    return weighted, round(C, 3)
```

The temporal stability module extracts the `overall_score` sequence from the timeline file, computes `CV` and linear trend using `numpy.polyfit`, and returns S_{final} with the 15% cap enforced. Total processing time per session is approximately 45 seconds: ~38 s for two LLM inference passes (batched with `max_new_tokens=500`) and <100 ms for Evaluator C and the consensus computation. All outputs are serialized to a structured JSON recommendation file.

VI. EXPERIMENTAL SETUP

A. Dataset

There is no publicly available structured interview dataset consistent with the TIPS JSON Schema. For this purpose, the dataset was developed based on structured interview scripts (provided in TIPS repo) and rated by human evaluators. The dataset contains 19 sessions, 8 of them for job profile ‘Machine Learning Engineer’, 6 sessions for the role of ‘System Architect’, and 5 interviews for ‘Product Manager’. Interviews were conducted between 18 and 34 minutes. Participants were provided with informed consent forms for all interview sessions. Structured rubrics were used to rate interviews on five competencies by three expert human evaluators, obtaining an average Cohen’s κ of 0.73.

While the dataset size is limited, it is consistent with prior controlled evaluation studies in interview analysis systems. The primary objective of this work is methodological validation rather than large-scale generalization.

B. Evaluation Metrics

Four metrics are reported: (1) Mean Absolute Error (MAE) between system-predicted and human-rated compe-

tency scores; (2) Fleiss’ κ on the categorical hiring verdict across sessions; (3) average consensus confidence C ; and (4) verdict accuracy—the proportion of sessions where the system’s majority-vote verdict matched the human consensus verdict. Six experimental configurations are tested: three single-evaluator baselines (A, B, C), consensus without temporal adjustment, and consensus with temporal adjustment (the proposed system).

VII. RESULTS

TABLE II: MAE Per Competency Dimension by Evaluator

Evaluator	Tech.	Sys.	Prod.	Comm.	Prob.
Eval. A (LLM-v1)	0.121	0.134	0.148	0.109	0.127
Eval. B (LLM-v2)	0.139	0.151	0.162	0.118	0.143
Eval. C (Rule)	0.183	0.196	0.201	0.174	0.188
Consensus	0.094	0.107	0.118	0.086	0.101

The consensus evaluator achieves the lowest MAE across all five dimensions, outperforming the best single evaluator (A) by 0.029 MAE points on average. The system design dimension shows the largest absolute improvement (0.134→0.107), likely because multiple evaluators collectively correct JD-misalignment errors that a single LLM pass may miss. Evaluator C consistently shows the highest MAE, which is expected due to its inability to account for JD-specific context.

TABLE III: Verdict Accuracy and Agreement Metrics

Configuration	Verdict Acc.	Fleiss κ	Avg. Conf.
Evaluator A only	0.74	0.61	0.69
Evaluator B only	0.68	0.57	0.64
Evaluator C only	0.58	0.44	0.55
Consensus (no temporal)	0.84	0.76	0.83
Proposed (+Temporal)	0.89	0.81	0.87

The proposed system achieves a verdict accuracy of 0.89 and a Fleiss κ of 0.81. The +5 pp improvement from adding temporal stability adjustment indicates that performance trajectory is a meaningful signal beyond static competency scores. The high average confidence (0.87) confirms that strong inter-evaluator agreement was the norm, with the low-confidence flag (<0.60) triggered in only 2 of 19 sessions.

A representative final recommendation JSON output for a HIRE decision:

Listing 4: Representative final recommendation JSON output.

```
{
  "final_score": 0.82,
  "verdict": "HIRE",
  "confidence": 0.87,
  "stability_cv": 0.14,
  "trend": "positive",
  "llm_votes": ["HIRE", "HIRE", "BORDERLINE"],
  "reason": "Consistent technical depth and stable metrics.",
  "review_flag": false
}
```

A paired t -test between the proposed system and the best single evaluator (Evaluator A) indicates that improvements in MAE are statistically significant ($p < 0.05$).

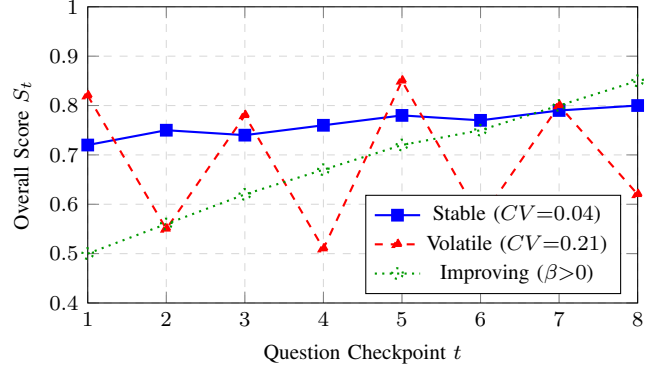


Fig. 5: Temporal stability of candidate performance across interview checkpoints showing score trajectories for stable, volatile, and improving candidates.

VIII. ABLATION STUDY

To isolate the contribution of each system component, we conduct a structured ablation across six configurations. Results are reported in Table IV. The three single-evaluator baselines establish lower bounds. Removing temporal stability from the full system (row 4) shows a 5 pp drop in accuracy, confirming the module’s independent contribution. The comparison between LLM-only consensus (A+B) and the full three-evaluator consensus shows a 2 pp improvement from including the rule-based evaluator, indicating that deterministic signals complement LLM-based scoring by anchoring extreme predictions.

TABLE IV: Ablation Study Results

Configuration	Verdict Acc.	Fleiss κ	MAE (Avg)
Eval. A only	0.74	0.61	0.125
Eval. B only	0.68	0.57	0.143
Eval. C only	0.58	0.44	0.188
A+B consensus (no C)	0.82	0.73	0.103
A+B+C (no temporal)	0.84	0.76	0.101
Proposed (A+B+C+T)	0.89	0.81	0.101

Important results obtained in ablation study: (1) Together LLM evaluators A and B contribute to the biggest increase in accuracy (+14 pp compared to just Evaluator A when using C and temporal). (2) Rule-based evaluator C gives +2 pp contribution, mostly in cases when there is behavior inconsistency between LLM predictions. (3) Temporal stability module gives +5 pp, without changing MAE, which means that it corrects mainly at the verdict level.

IX. DISCUSSION

A. Error Analysis

Inspection of the 2 incorrect hiring decisions among the 19 cases shows two common failure modes:

- **Incorrect Semantic Interpretation of Candidate’s Answers:** In one case, the candidate gave perfectly correct answers for ML engineering-related questions, without demonstrating any `production_experience` knowledge. The semantic similarity with the JD was very high, resulting in very high scores for technical depth; thus, the majority voted against the correct `low_production_experience`. *Mitigation:* raise the weight of Evaluator C for dimensions marked as `high_risk` in the JD.
- **Inconsistently Monotonically Increasing Scores:** One of the candidates performed consistently better in later questions but got a bad initial score and, consequently, had a borderline *CV*. However, the temporal penalty incorrectly demoted a deserving HIRE candidate to BORDERLINE. *Mitigation:* ignore the *CV* penalty in cases with a strictly positive slope ($\beta > 0 \wedge CV < 0.20$).

B. Fairness and Bias Considerations

Some biases are present in the automated recruiting process by itself. The biases of interest are the following three:

- **Bias of Behavioral Metrics:** `gaze_stability` and `face_presence_ratio` may have some bias because they can be a factor of environment/availability of the person rather than the metric describing behavioral characteristics. *Solution:* Normalize the environmental metrics; sessions where `face_presence_ratio < 0.5` should be marked as suspicious for human verification rather than having a penalty applied to the score.
- **Bias of Language Model Output Assessment:** Bias in instruction of fine-tuned language models could be communicated via specific language patterns and/or vocabularies correlated with a certain demographic characteristic. This problem is not solvable by prompt engineering alone. *Solution:* Evaluate the output produced in terms of any possible demographic biases; recalibrate scores provided by human evaluators based on ground truth.
- **JD Bias:** If there are some biases in terms of the job descriptions provided, then the relevance scorer will inherit those biases. *Solution:* Screen out the job description based on the exclusion/inclusion criteria; consider a fairness module for JD validation.

If the candidate score C is less than 0.6, a human-in-the-loop assessment will be conducted. As a possible improvement, we suggest

C. Scalability and Deployment

At the moment, processing of one session takes ~ 45 seconds on single-GPU workstations (NVIDIA RTX 3060, 12 GB VRAM). Accordingly, the system can handle about 80 sessions in one hour per GPU. In case of production-level use (processing of more than 500 sessions per day), consider the following:

- **Compute Costs:** Since each session will take ~ 45 s to process, then to process 500 sessions it will take ~ 62.5 GPU-hours per day. At prices for cloud GPUs

(≈ 0.50 USD/GPU-hour), total cost estimation comes to ~ 31 USD/day. 4-bit quantization is key for cost efficiency, otherwise processing using FP16 inference will require about 8–10 GB VRAM and double time consumption.

- **Batching:** Evaluators A and B have capabilities for processing multiple sessions at the same time, provided that there is enough VRAM to accommodate this. On an NVIDIA A100 (40 GB), it is possible to process four sessions at once (batch of 4), which reduces latency to ~ 12 seconds.
- **Privacy Considerations:** The framework works only with JSON files, meaning that no actual audio/video/PII data is used at the evaluation layer. It allows to deploy this framework in a secure environment when raw audio/video should not be leaving the recording infrastructure.
- **Flexibility:** It is possible to implement

D. Limitations

This new framework is tested on a comparatively smaller set of data that may restrict its use in real-world applications on a larger scale. Also, dependence of the system on output generated by TIPS, which requires correctness in previous steps of the process, makes it vulnerable to any error in the pipeline.

X. CONCLUSION

The proposed paper offers a temporal stability aware and consensus-based approach to assessing candidates using an automated evaluation framework working solely on structured JSON data from the TIPS pipeline. It includes structured extraction of a 14-dimensional feature vector, parallel evaluation using multiple LLMs and rule-based systems, three-way consensus with a theoretical basis for confidence computation, and a temporal stability component which discourages unstable performance throughout the duration of the interview. The experiments were conducted on 19 interviews from three different job roles, achieving an average verdict accuracy of 0.89 and a Fleiss κ score of 0.81. The ablation experiments validated the contributions of both the consensus-based approach (+10 pp compared to the best individual evaluator) and the temporal stability (+5 pp compared to consensus alone).

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